Proof. Let i be the smallest integer such that

$$d\left(V^{(i)}\right) = \alpha_p.$$

Then we also get $d\left(V'^{(i)}\right) = \alpha_p$. So

$$V'^{(i)} = V^{(i)} =: V^i.$$

Furthermore we have

$$V^{(i)} = F_k^+ \dots F_{n-1}^+ F_n^+ V^{(0)},$$
$$V'^{(i)} = F_k^+ \dots F_{n-1}^+ F_n^+ V'^{(0)}.$$

But both $V^{(i-1)}, \dots, V^{(0)}$ and $V'^{(i-1)}, \dots, V'^{(0)}$ have to be surjective at the appropriate vertices. This implies

$$\begin{aligned} &F_n^- F_{n-1}^- \dots F_k^- V^i \\ &\qquad = \begin{cases} F_n^- F_{n-1}^- \dots F_k^- F_k^+ \dots F_{n-1}^+ F_n^+ V^{(0)} = V^{(0)} = V, \\ F_n^- F_{n-1}^- \dots F_k^- F_k^+ \dots F_{n-1}^+ F_n^+ V'^{(0)} = V'^{(0)} = V'. \end{cases} \end{aligned}$$

□

These two corollaries show that there are only finitely many indecomposable representations (since there are only finitely many roots) and that the dimension vector of each of them is a positive root. The last statement of Gabriel's theorem follows from

Corollary 6.8.4. *For every positive root α, there is an indecomposable representation V with*

$$d(V) = \alpha.$$

Proof. Consider the sequence

$$s_n\alpha, s_{n-1}s_n\alpha, \dots.$$

Consider the first element of this sequence which is a negative root (this has to happen by Lemma 6.7.2) and look at one step before that, calling this element β. So β is a positive root and $s_i\beta$ is a negative root for some i. But since the s_i only change one coordinate, we get

$$\beta = \alpha_i$$

and

$$(s_q \dots s_{n-1}s_n)\alpha = \alpha_i.$$